Poincaré Compactification of a Two-Dimensional Hindmarsh–Rose-Type Polynomial System

We consider the two-dimensional Hindmarsh–Rose-type polynomial system

$$\boxed{\begin{aligned}\dot{x} &= y - ax^3 + bx^2 + I,\\ \dot{y} &= c - dx^2 - y,\end{aligned}} \tag{HR}$$

where a, b, c, d, I are real parameters. Throughout the generic cubic case, we assume

$$a \neq 0. \tag{1}$$

The purpose of this section is to perform the complete Poincaré compactification and to determine the singularities at infinity.

1 1. Standard Form and Polynomial Degree

Write the system in the standard form

$$\dot{x} = P(x, y), \qquad \dot{y} = Q(x, y), \tag{2}$$

where

$$\boxed{P(x, y) = y - ax^3 + bx^2 + I} \tag{3}$$

and

$$\boxed{Q(x, y) = c - dx^2 - y.} \tag{4}$$

The degree of P is

$$\deg P = 3 \tag{5}$$

provided $a \neq 0$, whereas

$$\deg Q = 2 \tag{6}$$

provided $d \neq 0$.

Hence the degree of the polynomial vector field is

$$\boxed{n = \max\{\deg P, \deg Q\} = 3.} \tag{1}$$

Therefore, in the Poincaré compactification, the appropriate time rescaling is

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\tau} = v^{n-1}\frac{\mathrm{d}}{\mathrm{d}t} = v^2\frac{\mathrm{d}}{\mathrm{d}t}.} \tag{2}$$

Equivalently,

$$\boxed{\frac{\mathrm{d}t}{\mathrm{d}\tau} = v^2.} \tag{7}$$

2 2. Homogeneous Components

We decompose P and Q into homogeneous components.

For P,

$$P(x,y) = P_3(x,y) + P_2(x,y) + P_1(x,y) + P_0(x,y), \tag{8}$$

where

$$P_3(x,y) = -ax^3, \tag{9}$$
$$P_2(x,y) = bx^2, \tag{10}$$
$$P_1(x,y) = y, \tag{11}$$
$$P_0(x,y) = I. \tag{12}$$

Thus

$$\boxed{P_3(x,y) = -ax^3.} \tag{3}$$

For Q,

$$Q(x,y) = Q_3(x,y) + Q_2(x,y) + Q_1(x,y) + Q_0(x,y), \tag{13}$$

where

$$Q_3(x,y) = 0, \tag{14}$$
$$Q_2(x,y) = -dx^2, \tag{15}$$
$$Q_1(x,y) = -y, \tag{16}$$
$$Q_0(x,y) = c. \tag{17}$$

Therefore

$$\boxed{Q_3(x,y) = 0.} \tag{4}$$

Consequently, the highest-degree homogeneous part of the vector field is

$$\boxed{(P_3, Q_3) = \left(-ax^3, 0\right).} \tag{5}$$

This highest-degree part determines the directions of the singularities at infinity.

3 3. Singular Directions at Infinity

For a polynomial planar system of degree n, the singular directions at infinity can be found by using the highest-degree homogeneous components P_n and Q_n.

In the x-chart, the singular directions satisfy

$$\boxed{Q_n(1,u) - uP_n(1,u) = 0.} \tag{6}$$

For the present system, $n = 3$, and therefore

$$P_3(1,u) = -a, \qquad Q_3(1,u) = 0. \tag{18}$$

Hence

$$Q_3(1,u) - uP_3(1,u) = 0 - u(-a) \tag{19}$$
$$= au. \tag{20}$$

Thus

$$au = 0. \tag{21}$$

Under the generic assumption $a \neq 0$,

$$\boxed{u = 0.} \tag{7}$$

Therefore one singular direction at infinity is obtained in the x-chart.
Similarly, in the y-chart, the singular directions satisfy

$$\boxed{P_n(u,1) - uQ_n(u,1) = 0.} \tag{8}$$

Since

$$P_3(u,1) = -au^3, \qquad Q_3(u,1) = 0, \tag{22}$$

we obtain

$$P_3(u,1) - uQ_3(u,1) = -au^3. \tag{23}$$

Thus

$$-au^3 = 0, \tag{24}$$

and hence

$$\boxed{u = 0.} \tag{9}$$

Therefore the generic system has two singular directions at infinity:

$$\boxed{[1:0:0] \qquad \text{and} \qquad [0:1:0].} \tag{10}$$

These correspond respectively to the x-axis and y-axis directions at infinity.

4 4. First Local Chart: the x-Chart

Consider the local chart defined by

$$\boxed{u = \frac{y}{x}, \qquad v = \frac{1}{x}.} \tag{11}$$

The inverse transformation is

$$\boxed{x = \frac{1}{v}, \qquad y = \frac{u}{v}.} \tag{12}$$

The coordinate v measures the approach to infinity:

$$|x| \to \infty \quad \Longleftrightarrow \quad v \to 0. \tag{25}$$

Thus the line

$$\boxed{v = 0} \tag{13}$$

represents the set of directions at infinity in this chart.
The variable

$$u = \frac{y}{x} \tag{26}$$

is the directional coordinate.

4.1 4.1 Transformation of the v-Equation

Since

$$x = \frac{1}{v}, \tag{27}$$

we have

$$\dot{x} = -\frac{\dot{v}}{v^2}. \tag{28}$$

Since

$$\dot{x} = P(x, y), \tag{29}$$

it follows that

$$-\frac{\dot{v}}{v^2} = P\left(\frac{1}{v}, \frac{u}{v}\right), \tag{30}$$

and consequently

$$\boxed{\dot{v} = -v^2 P\left(\frac{1}{v}, \frac{u}{v}\right).} \tag{14}$$

Now

$$P\left(\frac{1}{v}, \frac{u}{v}\right) = \frac{u}{v} - a\frac{1}{v^3} + b\frac{1}{v^2} + I \tag{31}$$

$$= \frac{-a + bv + uv^2 + Iv^3}{v^3}. \tag{32}$$

Therefore

$$\dot{v} = -v^2 \frac{-a + bv + uv^2 + Iv^3}{v^3} \tag{33}$$

$$= \frac{a - bv - uv^2 - Iv^3}{v}. \tag{34}$$

Thus

$$\boxed{\dot{v} = \frac{a - bv - uv^2 - Iv^3}{v}.} \tag{15}$$

4.2 4.2 Transformation of the u-Equation

Since

$$u = \frac{y}{x}, \tag{35}$$

we have

$$\dot{u} = \frac{x\dot{y} - y\dot{x}}{x^2}. \tag{36}$$

Using

$$x = \frac{1}{v}, \qquad y = \frac{u}{v}, \tag{37}$$

we obtain

$$\boxed{\dot{u} = v\left[Q\left(\frac{1}{v}, \frac{u}{v}\right) - uP\left(\frac{1}{v}, \frac{u}{v}\right)\right].} \tag{16}$$

First,

$$Q\left(\frac{1}{v}, \frac{u}{v}\right) = c - d\frac{1}{v^2} - \frac{u}{v} \tag{38}$$

$$= \frac{cv^3 - dv - uv^2}{v^3}. \tag{39}$$

Hence

$$Q - uP = \frac{cv^3 - dv - uv^2}{v^3} \tag{40}$$

$$- u\frac{-a + bv + uv^2 + Iv^3}{v^3} \tag{41}$$

$$= \frac{au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3}{v^3}. \tag{42}$$

Therefore,

$$\boxed{\dot{u} = \frac{au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3}{v^2}.} \tag{17}$$

4.3 4.3 Time Rescaling

Since the degree of the system is $n = 3$, we use

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\tau} = v^2 \frac{\mathrm{d}}{\mathrm{d}t}.} \tag{18}$$

Hence

$$u' = v^2\dot{u}, \qquad v' = v^2\dot{v}, \tag{43}$$

where the prime denotes differentiation with respect to τ.

From (15),

$$v' = v^2\frac{a - bv - uv^2 - Iv^3}{v} \tag{44}$$

$$= av - bv^2 - uv^3 - Iv^4. \tag{45}$$

From (17),

$$u' = v^2\frac{au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3}{v^2} \tag{46}$$

$$= au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3. \tag{47}$$

Thus the compactified vector field in the x-chart is

$$\boxed{\begin{aligned} u' &= au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3, \\ v' &= av - bv^2 - uv^3 - Iv^4. \end{aligned}} \tag{19}$$

This system is regular at $v = 0$.

4.4 4.4 Dynamics on the Line at Infinity

Setting

$$v = 0 \tag{48}$$

in (19), we obtain

$$\boxed{u' = au, \qquad v' = 0.} \tag{20}$$

Therefore the singularity condition is

$$au = 0. \tag{49}$$

For $a \neq 0$,

$$\boxed{u_* = 0.} \tag{21}$$

Thus the singular point at infinity in the x-chart is

$$\boxed{p_x = (0, 0).} \tag{22}$$

Since

$$u = \frac{y}{x}, \tag{50}$$

the condition $u = 0$ means

$$\frac{y}{x} \to 0. \tag{51}$$

Therefore

$$\boxed{p_x \text{ represents the } x\text{-axis direction at infinity.}} \tag{23}$$

In homogeneous projective coordinates,

$$\boxed{p_x = [1 : 0 : 0].} \tag{24}$$

4.5 4.5 Jacobian at p_x

Define

$$f(u, v) = au - (d + bu)v - u(1 + u)v^2 + (c - Iu)v^3, \tag{52}$$
$$g(u, v) = av - bv^2 - uv^3 - Iv^4. \tag{53}$$

The Jacobian matrix is

$$J_x(u, v) = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}. \tag{54}$$

We have

$$f_u = a - bv - (1 + 2u)v^2 - Iv^3, \tag{55}$$
$$f_v = -(d + bu) - 2u(1 + u)v + 3(c - Iu)v^2, \tag{56}$$
$$g_u = -v^3, \tag{57}$$
$$g_v = a - 2bv - 3uv^2 - 4Iv^3. \tag{58}$$

Therefore

$$\boxed{J_x(u, v) = \begin{pmatrix} a - bv - (1 + 2u)v^2 - Iv^3 & -(d + bu) - 2u(1 + u)v + 3(c - Iu)v^2 \\ -v^3 & a - 2bv - 3uv^2 - 4Iv^3 \end{pmatrix}.} \tag{25}$$

At $p_x = (0, 0)$,

$$\boxed{J_x(p_x) = \begin{pmatrix} a & -d \\ 0 & a \end{pmatrix}.} \tag{26}$$

The characteristic equation is

$$\det \begin{pmatrix} a - \lambda & -d \\ 0 & a - \lambda \end{pmatrix} = 0, \tag{59}$$

which gives

$$(a - \lambda)^2 = 0. \tag{60}$$

Hence

$$\boxed{\lambda_1 = \lambda_2 = a.} \tag{27}$$

4.6 4.6 Classification of p_x

If

$$a > 0, \tag{61}$$

then

$$\lambda_1 = \lambda_2 = a > 0, \tag{62}$$

so p_x is unstable.

For $d \neq 0$, the matrix

$$\begin{pmatrix} a & -d \\ 0 & a \end{pmatrix} \tag{63}$$

has only one eigenvector and therefore has a nontrivial Jordan block. Consequently, p_x is generically an unstable improper node.

Thus,

$$\boxed{a > 0,\ d \neq 0 \quad \Longrightarrow \quad p_x \text{ is an unstable improper node.}} \tag{28}$$

If $d = 0$, then

$$J_x(p_x) = aI, \tag{64}$$

and the point is a star node.

Similarly, if

$$a < 0, \tag{65}$$

then

$$\lambda_1 = \lambda_2 = a < 0, \tag{66}$$

and p_x is stable.

For $d \neq 0$,

$$\boxed{a < 0,\ d \neq 0 \quad \Longrightarrow \quad p_x \text{ is a stable improper node.}} \tag{29}$$

For $d = 0$, it is a stable star node.

Therefore, for $a \neq 0$, the point p_x is hyperbolic.

5 5. Second Local Chart: the y-Chart

The first chart does not cover the vertical direction. We therefore introduce the second local chart

$$\boxed{u = \frac{x}{y}, \qquad v = \frac{1}{y}.} \tag{30}$$

The inverse transformation is

$$\boxed{x = \frac{u}{v}, \qquad y = \frac{1}{v}.} \tag{31}$$

Again,

$$v = 0 \tag{67}$$

represents infinity in this chart, while $u = x/y$ represents direction.

5.1 5.1 Transformation of the v-Equation

Since

$$y = \frac{1}{v}, \tag{68}$$

we have

$$\dot{y} = -\frac{\dot{v}}{v^2}. \tag{69}$$

Since $\dot{y} = Q(x, y)$,

$$-\frac{\dot{v}}{v^2} = Q\left(\frac{u}{v}, \frac{1}{v}\right), \tag{70}$$

so

$$\boxed{\dot{v} = -v^2 Q\left(\frac{u}{v}, \frac{1}{v}\right).} \tag{32}$$

We have

$$Q\left(\frac{u}{v}, \frac{1}{v}\right) = c - d\frac{u^2}{v^2} - \frac{1}{v} \tag{71}$$

$$= \frac{cv^3 - du^2 v - v^2}{v^3}. \tag{72}$$

Therefore

$$\dot{v} = -v^2 \frac{cv^3 - du^2 v - v^2}{v^3} \tag{73}$$

$$= \frac{du^2 v + v^2 - cv^3}{v}. \tag{74}$$

Hence

$$\boxed{\dot{v} = \frac{du^2 v + v^2 - cv^3}{v}.} \tag{33}$$

5.2 5.2 Transformation of the u-Equation

Since

$$u = \frac{x}{y}, \tag{75}$$

we have

$$\dot{u} = \frac{y\dot{x} - x\dot{y}}{y^2}. \tag{76}$$

Using

$$x = \frac{u}{v}, \qquad y = \frac{1}{v}, \tag{77}$$

we obtain

$$\boxed{\dot{u} = v\left[P\left(\frac{u}{v}, \frac{1}{v}\right) - uQ\left(\frac{u}{v}, \frac{1}{v}\right)\right].} \tag{34}$$

First,

$$P\left(\frac{u}{v}, \frac{1}{v}\right) = \frac{1}{v} - a\frac{u^3}{v^3} + b\frac{u^2}{v^2} + I \tag{78}$$

$$= \frac{-au^3 + bu^2 v + uv^2 + Iv^3}{v^3}. \tag{79}$$

Also,

$$Q\left(\frac{u}{v},\frac{1}{v}\right)=\frac{cv^3-du^2v-v^2}{v^3}. \tag{80}$$

Therefore

$$P-uQ=\frac{-au^3+(bu^2+du^3)v+(u+u^2)v^2+(I-cu)v^3}{v^3}. \tag{81}$$

Consequently,

$$\boxed{\dot{u}=\frac{-au^3+(bu^2+du^3)v+(u+u^2)v^2+(I-cu)v^3}{v^2}.} \tag{35}$$

5.3 5.3 Time Rescaling

Again, since $n=3$,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}=v^2\frac{\mathrm{d}}{\mathrm{d}t}. \tag{82}$$

Thus

$$u'=v^2\dot{u},\qquad v'=v^2\dot{v}. \tag{83}$$

From (35),

$$u'=-au^3+(bu^2+du^3)v+(u+u^2)v^2+(I-cu)v^3. \tag{84}$$

From (33),

$$v'=v^2\frac{du^2v+v^2-cv^3}{v} \tag{85}$$

$$=du^2v^2+v^3-cv^4. \tag{86}$$

Therefore the compactified system in the y-chart is

$$\boxed{\begin{aligned}u'&=-au^3+(bu^2+du^3)v+(u+u^2)v^2+(I-cu)v^3,\\ v'&=du^2v^2+v^3-cv^4.\end{aligned}} \tag{36}$$

5.4 5.4 Dynamics on the Line at Infinity

Setting

$$v=0, \tag{87}$$

we obtain

$$\boxed{u'=-au^3,\qquad v'=0.} \tag{37}$$

Hence the singularity condition is

$$-au^3=0. \tag{88}$$

For $a\neq 0$,

$$\boxed{u_*=0.} \tag{38}$$

Therefore the singular point in the y-chart is

$$\boxed{p_y=(0,0).} \tag{39}$$

Since

$$u = \frac{x}{y}, \tag{89}$$

the condition $u = 0$ corresponds to

$$x = 0. \tag{90}$$

Hence

$$\boxed{p_y \text{ represents the } y\text{-axis direction at infinity.}} \tag{40}$$

In homogeneous projective coordinates,

$$\boxed{p_y = [0 : 1 : 0].} \tag{41}$$

5.5 5.5 Jacobian at p_y

Define

$$f(u, v) = -au^3 + (bu^2 + du^3)v + (u + u^2)v^2 + (I - cu)v^3, \tag{91}$$

$$g(u, v) = du^2v^2 + v^3 - cv^4. \tag{92}$$

Every term in f and g has total degree at least 2. Consequently, there are no linear terms in the vector field at the origin.

Hence

$$\boxed{J_y(p_y) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.} \tag{42}$$

Therefore the eigenvalues are

$$\boxed{\lambda_1 = \lambda_2 = 0.} \tag{43}$$

Thus p_y is a nonhyperbolic, degenerate singularity.

$$\boxed{p_y \text{ is nonhyperbolic and cannot be classified by linearization alone.}} \tag{44}$$

Higher-order terms must be considered in order to determine its local phase portrait.

6 6. Why the Two Infinity Points Are Different

The singularity p_x corresponds to

$$y = 0, \qquad |x| \to \infty, \tag{93}$$

whereas p_y corresponds to

$$x = 0, \qquad |y| \to \infty. \tag{94}$$

Therefore they represent different directions at infinity:

$$\boxed{p_x = [1 : 0 : 0], \qquad p_y = [0 : 1 : 0].} \tag{95}$$

The asymmetry between the two singularities is caused by the highest degree part

$$(P_3, Q_3) = (-ax^3, 0). \tag{96}$$

At the x-axis direction,

$$y = 0, \tag{97}$$

and the cubic term

$$-ax^3 \tag{98}$$

is nonzero.

By contrast, along the y-axis direction,

$$x = 0, \tag{99}$$

and therefore

$$P_3(0, y) = 0. \tag{100}$$

This explains why the x-axis infinity point has a nonzero linearization, whereas the y-axis infinity point is degenerate.

7 7. Transition Between the Two Charts

The two local charts are related by the transition map.

In the x-chart,

$$u_1 = \frac{y}{x}, \qquad v_1 = \frac{1}{x}. \tag{101}$$

In the y-chart,

$$u_2 = \frac{x}{y}, \qquad v_2 = \frac{1}{y}. \tag{102}$$

Therefore,

$$u_2 = \frac{x}{y} = \frac{1}{y/x} = \frac{1}{u_1}, \tag{103}$$

$$v_2 = \frac{1}{y} = \frac{1/x}{y/x} = \frac{v_1}{u_1}. \tag{104}$$

Hence

$$\boxed{u_2 = \frac{1}{u_1}, \qquad v_2 = \frac{v_1}{u_1}.} \tag{45}$$

This transition map is defined on the overlap where

$$u_1 \neq 0. \tag{105}$$

The point $u_1 = 0$ is the horizontal direction, which is not represented by the y-chart.

Similarly, $u_2 = 0$ represents the vertical direction, which is not represented by the x-chart.

8 8. Important Parameter Dependence

The locations of the singular directions at infinity are determined only by the highest-degree homogeneous terms

$$P_3(x, y) = -ax^3, \qquad Q_3(x, y) = 0. \tag{106}$$

Consequently, for $a \neq 0$, the singular directions

$$[1 : 0 : 0], \qquad [0 : 1 : 0] \tag{107}$$

do not depend on b, c, d, or I.

However, the detailed local dynamics near the singularities may depend on the lower-degree terms.

For example,

$$J_x(p_x) = \begin{pmatrix} a & -d \\ 0 & a \end{pmatrix}, \tag{108}$$

so d affects the eigenvector and Jordan structure even though it does not affect the location of the singular direction.

Thus the general principle is

$$\boxed{\text{highest-degree terms} \quad \Longrightarrow \quad \text{singular directions at infinity,}} \tag{46}$$

whereas

$$\boxed{\text{all polynomial terms} \quad \Longrightarrow \quad \text{local dynamics near infinity.}} \tag{47}$$

9 9. Degenerate Case $a = 0$

The preceding calculations assume

$$a \neq 0. \tag{109}$$

If

$$a = 0, \tag{110}$$

then the cubic term disappears and the system becomes

$$\dot{x} = y + bx^2 + I, \qquad \dot{y} = c - dx^2 - y. \tag{111}$$

The degree is then generically

$$\boxed{n = 2,} \tag{112}$$

provided at least one quadratic coefficient is nonzero.

Consequently, the appropriate Poincaré time rescaling is no longer v^2, but

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\tau} = v\frac{\mathrm{d}}{\mathrm{d}t}.} \tag{113}$$

Thus, when parameters cause the degree of the polynomial vector field to change, the compactification must be recomputed using the actual degree of the resulting system.

10 10. Reusable Procedure for a General Polynomial System

For a general planar polynomial system

$$\dot{x} = P(x, y), \qquad \dot{y} = Q(x, y), \tag{114}$$

the following procedure can be applied systematically.

Step 1: Determine the degree

Set

$$n = \max\{\deg P, \deg Q\}. \tag{115}$$

Step 2: Extract the highest-degree homogeneous terms

Write

$$P = P_n + \cdots, \qquad Q = Q_n + \cdots. \tag{116}$$

Step 3: Find the singular directions at infinity

In the x-chart solve

$$\boxed{Q_n(1,u) - uP_n(1,u) = 0.} \tag{117}$$

In the y-chart solve

$$\boxed{P_n(u,1) - uQ_n(u,1) = 0.} \tag{118}$$

Step 4: Introduce the local charts

For the x-chart,

$$\boxed{x = \frac{1}{v}, \qquad y = \frac{u}{v}.} \tag{119}$$

For the y-chart,

$$\boxed{x = \frac{u}{v}, \qquad y = \frac{1}{v}.} \tag{120}$$

Step 5: Transform the vector field

In the x-chart,

$$\boxed{\dot{u} = v(Q - uP), \qquad \dot{v} = -v^2 P,} \tag{121}$$

where P and Q are evaluated at

$$\left(\frac{1}{v}, \frac{u}{v}\right). \tag{122}$$

In the y-chart,

$$\boxed{\dot{u} = v(P - uQ), \qquad \dot{v} = -v^2 Q,} \tag{123}$$

where P and Q are evaluated at

$$\left(\frac{u}{v}, \frac{1}{v}\right). \tag{124}$$

Step 6: Rescale time

Use

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\tau} = v^{n-1}\frac{\mathrm{d}}{\mathrm{d}t}.} \tag{125}$$

Step 7: Restrict to infinity

Set

$$v = 0. \tag{126}$$

Step 8: Find and classify the singularities

Solve the resulting equation for u, then calculate the Jacobian matrix of the compactified vector field.

If the eigenvalues have nonzero real parts, the singularity is hyperbolic and the local phase portrait is determined by the linearization.

If zero eigenvalues occur, the singularity is nonhyperbolic and higher-order analysis is required.

11 11. Final Results for the Hindmarsh–Rose-Type System

For the system

$$\boxed{\begin{aligned}\dot{x} &= y - ax^3 + bx^2 + I,\\ \dot{y} &= c - dx^2 - y,\end{aligned}} \tag{127}$$

with $a \neq 0$, the degree is

$$\boxed{n = 3}. \tag{128}$$

The highest-degree homogeneous part is

$$\boxed{(P_3, Q_3) = (-ax^3, 0).} \tag{129}$$

The singular directions at infinity are

$$\boxed{[1:0:0] \quad \text{and} \quad [0:1:0].} \tag{130}$$

The point

$$\boxed{p_x = [1:0:0]} \tag{131}$$

is represented in the x-chart by

$$(u, v) = (0, 0) \tag{132}$$

and has Jacobian

$$\boxed{J_x(p_x) = \begin{pmatrix} a & -d \\ 0 & a \end{pmatrix}.} \tag{133}$$

Its eigenvalues are

$$\boxed{\lambda_1 = \lambda_2 = a.} \tag{134}$$

Thus, for $a \neq 0$,

$$\boxed{\begin{cases} a > 0: & p_x \text{ is an unstable node,} \\ a < 0: & p_x \text{ is a stable node.} \end{cases}} \tag{135}$$

For $d \neq 0$, the node is generically an improper node.

The point

$$\boxed{p_y = [0:1:0]} \tag{136}$$

is represented in the y-chart by

$$(u, v) = (0, 0) \tag{137}$$

and has Jacobian

$$\boxed{J_y(p_y) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.} \tag{138}$$

Hence

$$\boxed{\lambda_1 = \lambda_2 = 0,} \tag{139}$$

so

$$\boxed{p_y \text{ is a degenerate nonhyperbolic singularity at infinity.}} \tag{140}$$

Its local phase portrait cannot be obtained from the linearization and requires an analysis of the higher-order terms.

12 1. Equilibrium Equations

An equilibrium point

$$E^* = (x^*, y^*) \tag{141}$$

satisfies

$$\dot{x} = 0, \qquad \dot{y} = 0. \tag{142}$$

Therefore,

$$y^* - a(x^*)^3 + b(x^*)^2 + I = 0, \tag{1}$$

and

$$c - d(x^*)^2 - y^* = 0. \tag{2}$$

From the second equation,

$$\boxed{y^* = c - d(x^*)^2.} \tag{3}$$

Substituting (3) into the first equilibrium equation gives

$$c - d(x^*)^2 - a(x^*)^3 + b(x^*)^2 + I = 0. \tag{143}$$

Hence

$$-a(x^*)^3 + (b-d)(x^*)^2 + (c+I) = 0. \tag{144}$$

Multiplying by -1 gives

$$\boxed{a(x^*)^3 + (d-b)(x^*)^2 - (c+I) = 0.} \tag{4}$$

Thus all finite equilibria are obtained by solving the cubic equation (4), and then using (3) to obtain their y-coordinates.

For convenience, define

$$\boxed{B = d - b, \qquad S = c + I.} \tag{5}$$

Then the equilibrium equation becomes

$$\boxed{ax^3 + Bx^2 - S = 0.} \tag{6}$$

Hence every equilibrium is of the form

$$\boxed{E^* = \left(x^*, c - d(x^*)^2\right),} \tag{7}$$

where x^* is a real root of (6).

13 2. Number of Equilibria

Assume

$$a \neq 0. \tag{145}$$

Then (6) is a cubic equation. Its discriminant is

$$\boxed{\mathcal{D} = 4B^3S - 27a^2S^2.} \tag{8}$$

Returning to the original parameters,

$$\boxed{\mathcal{D} = 4(d-b)^3(c+I) - 27a^2(c+I)^2.} \tag{9}$$

The number of distinct real equilibria is determined by the sign of $\mathcal{D}$:

$\mathcal{D} > 0$	three distinct real equilibria,
$\mathcal{D} = 0$	at least one multiple real equilibrium,
$\mathcal{D} < 0$	one real equilibrium.

(10)

Thus the discriminant provides a direct criterion for the number of finite equilibria.

14 3. Explicit Solution of the Cubic

Starting from

$$ax^3 + Bx^2 - S = 0, \tag{146}$$

divide by a:

$$x^3 + \frac{B}{a}x^2 - \frac{S}{a} = 0. \tag{11}$$

Introduce the change of variable

$$\boxed{x = z - \frac{B}{3a}.} \tag{12}$$

This removes the quadratic term and yields the depressed cubic

$$z^3 + pz + q = 0, \tag{13}$$

where

$$\boxed{p = -\frac{B^2}{3a^2},} \tag{14}$$

and

$$\boxed{q = \frac{2B^3}{27a^3} - \frac{S}{a}.} \tag{15}$$

The Cardano quantity is

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3. \tag{16}$$

After simplification,

$$\boxed{\Delta = \frac{27a^2S^2 - 4B^3S}{108a^4}.} \tag{17}$$

In terms of the cubic discriminant,

$$\boxed{\mathcal{D} = -108a^4\Delta.} \tag{18}$$

Therefore,

$$\mathcal{D} > 0 \quad \Longleftrightarrow \quad \Delta < 0, \tag{147}$$

which corresponds to three distinct real roots.

15 4. The One-Real-Equilibrium Case

When

$$\Delta > 0 \tag{148}$$

or equivalently

$$\mathcal{D} < 0, \tag{149}$$

there is one real root.

By Cardano's formula,

$$z = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}. \tag{150}$$

Consequently,

$$\boxed{x^* = -\frac{B}{3a} + \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}.} \tag{19}$$

Using

$$-\frac{q}{2} = \frac{S}{2a} - \frac{B^3}{27a^3}, \tag{151}$$

we obtain

$$\boxed{x^* = -\frac{B}{3a} + \sqrt[3]{\frac{S}{2a} - \frac{B^3}{27a^3} + \sqrt{\Delta}} + \sqrt[3]{\frac{S}{2a} - \frac{B^3}{27a^3} - \sqrt{\Delta}}.} \tag{20}$$

The corresponding equilibrium is

$$\boxed{E^* = \left(x^*, c - d(x^*)^2\right).} \tag{21}$$

16 5. The Three-Real-Equilibria Case

When

$$\mathcal{D} > 0, \tag{152}$$

the cubic has three distinct real roots.

Since

$$p = -\frac{B^2}{3a^2} < 0 \tag{153}$$

for $B \neq 0$, the trigonometric form of the cubic solution is convenient.

Define

$$\boxed{\theta = \frac{1}{3}\arccos\left(1 - \frac{27a^2S}{2B^3}\right).} \tag{22}$$

Then the three real roots may be written as

$$\boxed{x_k = -\frac{B}{3a} + \frac{2|B|}{3|a|}\cos\left(\theta - \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.} \tag{23}$$

The corresponding equilibria are

$$\boxed{E_k = \left(x_k, c - dx_k^2\right), \qquad k = 0, 1, 2.} \tag{24}$$

For numerical calculations, it is often more convenient to obtain the roots using the numerical polynomial-root algorithm rather than the explicit trigonometric formula. Nevertheless, formula (23) gives an analytical representation of all three roots.

17 6. Special Case $c + I = 0$

Suppose

$$S = c + I = 0. \tag{154}$$

Then the equilibrium equation becomes

$$ax^3 + Bx^2 = 0. \tag{155}$$

Factoring,

$$x^2(ax + B) = 0. \tag{156}$$

Hence

$$\boxed{x_1^* = 0} \tag{25}$$

is a double root, while

$$\boxed{x_2^* = -\frac{B}{a} = \frac{b-d}{a}.} \tag{26}$$

The corresponding y-coordinates are

$$y_1^* = c, \tag{157}$$

and

$$y_2^* = c - d\left(\frac{b-d}{a}\right)^2. \tag{158}$$

Therefore the distinct equilibria are

$$\boxed{E_1 = (0, c),} \tag{27}$$

and

$$\boxed{E_2 = \left(\frac{b-d}{a},\, c - d\frac{(b-d)^2}{a^2}\right).} \tag{28}$$

The root $x = 0$ is algebraically double and therefore corresponds to a degenerate parameter configuration.

18 7. Jacobian Matrix at an Arbitrary Equilibrium

The vector field is

$$F(x, y) = \begin{pmatrix} y - ax^3 + bx^2 + I \\ c - dx^2 - y \end{pmatrix}. \tag{159}$$

The Jacobian matrix is

$$DF(x, y) = \begin{pmatrix} \dfrac{\partial \dot{x}}{\partial x} & \dfrac{\partial \dot{x}}{\partial y} \\ \dfrac{\partial \dot{y}}{\partial x} & \dfrac{\partial \dot{y}}{\partial y} \end{pmatrix}. \tag{160}$$

Hence

$$\boxed{DF(x, y) = \begin{pmatrix} -3ax^2 + 2bx & 1 \\ -2dx & -1 \end{pmatrix}.} \tag{29}$$

At an equilibrium

$$E^* = (x^*, y^*), \tag{161}$$

the Jacobian is therefore

$$\boxed{J^* = \begin{pmatrix} -3a(x^*)^2 + 2bx^* & 1 \\ -2dx^* & -1 \end{pmatrix}.} \tag{30}$$

19 8. Trace and Determinant

The trace of J^* is

$$\operatorname{tr}(J^*) = -3a(x^*)^2 + 2bx^* - 1. \tag{162}$$

Thus

$$\boxed{T(x^*) = -3a(x^*)^2 + 2bx^* - 1.} \tag{31}$$

The determinant is

$$(J^*) = \left[-3a(x^*)^2 + 2bx^*\right](-1) - (1)(-2dx^*) \tag{163}$$

$$= 3a(x^*)^2 - 2bx^* + 2dx^*. \tag{164}$$

Therefore

$$\boxed{D(x^*) = 3a(x^*)^2 + 2(d-b)x^*.} \tag{32}$$

Using $B = d - b$,

$$\boxed{D(x^*) = x^*\left(3ax^* + 2B\right).} \tag{33}$$

20 9. Eigenvalues

The characteristic polynomial of J^* is

$$\lambda^2 - T\lambda + D = 0. \tag{34}$$

Consequently,

$$\boxed{\lambda_\pm = \frac{1}{2}\left[T \pm \sqrt{T^2 - 4D}\right].} \tag{35}$$

Here

$$T = -3a(x^*)^2 + 2bx^* - 1 \tag{165}$$

and

$$D = x^*\left[3ax^* + 2(d-b)\right]. \tag{166}$$

21 10. Classification of Equilibria

The local type of an equilibrium can be determined from its trace, determinant, and discriminant.

10.1 Saddle

If

$$\boxed{D < 0,} \tag{36}$$

then the eigenvalues have opposite signs. Therefore

$$\boxed{E^* \text{ is a saddle.}} \tag{37}$$

10.2 Stable Equilibrium

If

$$D > 0 \tag{167}$$

and

$$T < 0, \tag{168}$$

then both eigenvalues have negative real parts.

Therefore

$$\boxed{E^* \text{ is locally asymptotically stable.}} \tag{38}$$

More precisely:

$$T^2 - 4D > 0 \quad \Longrightarrow \quad \text{stable node,} \tag{169}$$

whereas

$$T^2 - 4D < 0 \quad \Longrightarrow \quad \text{stable focus.} \tag{170}$$

10.3 Unstable Equilibrium

If

$$D > 0 \tag{171}$$

and

$$T > 0, \tag{172}$$

then both eigenvalues have positive real parts.

Hence

$$\boxed{E^* \text{ is unstable.}} \tag{39}$$

More precisely,

$$T^2 - 4D > 0 \quad \Longrightarrow \quad \text{unstable node}, \tag{173}$$

while

$$T^2 - 4D < 0 \quad \Longrightarrow \quad \text{unstable focus}. \tag{174}$$

22 11. Hopf Bifurcation Condition

For a planar system, a Hopf bifurcation requires a pair of purely imaginary eigenvalues. For the characteristic polynomial

$$\lambda^2 - T\lambda + D = 0, \tag{175}$$

the necessary linear conditions are

$$\boxed{T = 0, \qquad D > 0.} \tag{40}$$

For the present system,

$$T(x^*) = -3a(x^*)^2 + 2bx^* - 1. \tag{176}$$

Thus the trace condition is

$$-3a(x^*)^2 + 2bx^* - 1 = 0, \tag{177}$$

or

$$\boxed{3a(x^*)^2 - 2bx^* + 1 = 0.} \tag{41}$$

This equation determines the candidate equilibrium locations at Hopf.

23 12. Hopf Equilibrium Coordinates

Solving

$$3a(x^*)^2 - 2bx^* + 1 = 0 \tag{178}$$

gives

$$\boxed{x_H^{\pm} = \frac{b \pm \sqrt{b^2 - 3a}}{3a}.} \tag{42}$$

Therefore real Hopf candidates require

$$\boxed{b^2 - 3a \geq 0.} \tag{43}$$

For each candidate $x_H^{\pm}$, the corresponding equilibrium y-coordinate is

$$\boxed{y_H^{\pm} = c - d(x_H^{\pm})^2.} \tag{44}$$

Thus the candidate Hopf equilibria are

$$\boxed{E_H^{\pm} = \left(x_H^{\pm}, c - d(x_H^{\pm})^2\right).} \tag{45}$$

24 13. Determinant at the Hopf Point

At a Hopf candidate,

$$x_H^{\pm} = \frac{b \pm \sqrt{b^2 - 3a}}{3a}. \tag{179}$$

The determinant is

$$D_H = x_H \left[3ax_H + 2(d-b)\right]. \tag{46}$$

Since

$$3ax_H^{\pm} = b \pm \sqrt{b^2 - 3a}, \tag{180}$$

we obtain

$$3ax_H^{\pm} + 2(d-b) = b \pm \sqrt{b^2 - 3a} + 2d - 2b \tag{181}$$

$$= 2d - b \pm \sqrt{b^2 - 3a}. \tag{182}$$

Therefore

$$\boxed{D_H^{\pm} = x_H^{\pm}\left[2d - b \pm \sqrt{b^2 - 3a}\right].} \tag{47}$$

For a Hopf bifurcation candidate, we require

$$\boxed{D_H^{\pm} > 0.} \tag{48}$$

At such a point,

$$\lambda_{\pm} = \pm i\omega_H, \tag{183}$$

where

$$\boxed{\omega_H = \sqrt{D_H}.} \tag{49}$$

25 14. Hopf Parameter Condition

The equilibrium equation is

$$ax^3 + (d-b)x^2 - (c+I) = 0. \tag{184}$$

Therefore, at a Hopf point $x = x_H$, we must have

$$ax_H^3 + (d-b)x_H^2 - (c+I_H) = 0. \tag{185}$$

Hence

$$\boxed{c + I_H = ax_H^3 + (d-b)x_H^2.} \tag{50}$$

Consequently,

$$\boxed{I_H = ax_H^3 + (d-b)x_H^2 - c.} \tag{51}$$

For the two Hopf candidates $x_H^{\pm}$,

$$\boxed{I_H^{\pm} = a(x_H^{\pm})^3 + (d-b)(x_H^{\pm})^2 - c.} \tag{52}$$

Thus the Hopf bifurcation candidates can be parameterized as

$$\boxed{x_H^{\pm} = \frac{b \pm \sqrt{b^2 - 3a}}{3a},} \tag{53}$$

and

$$\boxed{I_H^{\pm} = a(x_H^{\pm})^3 + (d-b)(x_H^{\pm})^2 - c.} \tag{54}$$

26 15. Simplification of the Hopf Parameter Relation

At a Hopf point,

$$3ax_H^2 - 2bx_H + 1 = 0. \tag{186}$$

Hence

$$3ax_H^2 = 2bx_H - 1. \tag{55}$$

Multiplying by $x_H/3$,

$$ax_H^3 = \frac{x_H(2bx_H - 1)}{3}. \tag{56}$$

Therefore

$$\begin{aligned} c + I_H &= ax_H^3 + (d-b)x_H^2 && (187)\\ &= \frac{2b}{3}x_H^2 - \frac{x_H}{3} + (d-b)x_H^2 && (188)\\ &= \left(d - \frac{b}{3}\right)x_H^2 - \frac{x_H}{3}. && (189) \end{aligned}$$

Consequently,

$$\boxed{I_H = \left(d - \frac{b}{3}\right)x_H^2 - \frac{x_H}{3} - c.} \tag{57}$$

This form is often convenient for algebraic simplification.

27 16. Relation to Limit Cycles

The existence of a Hopf candidate does not by itself prove the existence of a periodic orbit.

The logical sequence is

$$\boxed{\text{equilibrium} \quad \longrightarrow \quad T = 0,\ D > 0 \quad \longrightarrow \quad \text{Hopf candidate.}} \tag{58}$$

For a generic Hopf bifurcation, one additionally needs the appropriate nondegeneracy and transversality conditions.

In particular, the eigenvalues must cross the imaginary axis as the bifurcation parameter varies, and the first Lyapunov coefficient must satisfy

$$\boxed{l_1 \neq 0.} \tag{59}$$

When the generic Hopf hypotheses are satisfied, a small-amplitude limit cycle is created or destroyed at the Hopf bifurcation.

Thus

$$\boxed{T = 0,\ D > 0} \tag{190}$$

provides only the linear Hopf condition, whereas

$$\boxed{l_1 \neq 0} \tag{191}$$

is needed to establish a generic nondegenerate Hopf bifurcation.

28 17. Divergence and a No-Limit-Cycle Criterion

The divergence of the vector field is

$$\nabla \cdot F = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} \tag{192}$$

$$= (-3ax^2 + 2bx) + (-1). \tag{193}$$

Therefore

$$\boxed{\nabla \cdot F = -3ax^2 + 2bx - 1.} \tag{60}$$

Notice that this is precisely the trace of the Jacobian:

$$\boxed{\nabla \cdot F = T(x).} \tag{61}$$

Assume

$$a > 0. \tag{194}$$

The divergence is then a concave quadratic function of x:

$$\nabla \cdot F = -3ax^2 + 2bx - 1. \tag{195}$$

Its maximum occurs at

$$x = \frac{b}{3a}. \tag{196}$$

The maximum value is

$$\max_x \nabla \cdot F = -3a\left(\frac{b}{3a}\right)^2 + 2b\left(\frac{b}{3a}\right) - 1 \tag{197}$$

$$= -\frac{b^2}{3a} + \frac{2b^2}{3a} - 1 \tag{198}$$

$$= \frac{b^2}{3a} - 1 \tag{199}$$

$$= \frac{b^2 - 3a}{3a}. \tag{200}$$

Consequently, if

$$\boxed{a > 0, \qquad b^2 \le 3a,} \tag{62}$$

then

$$\boxed{\nabla \cdot F \le 0 \qquad \text{for all } (x, y).} \tag{63}$$

Since the divergence is not identically zero, Bendixson's criterion rules out nonconstant periodic orbits in the plane.

Therefore,

$$\boxed{a > 0, \qquad b^2 \le 3a \quad \Longrightarrow \quad \text{no planar limit cycle.}} \tag{64}$$

This result is also consistent with the Hopf condition. Indeed, for

$$a > 0, \qquad b^2 < 3a, \tag{201}$$

the equation

$$3ax^2 - 2bx + 1 = 0 \tag{202}$$

has no real solution, so no Hopf candidate exists.

29 18. Practical Procedure for a Concrete Parameter Set

Suppose that numerical values of

$$(a, b, c, d, I) \tag{203}$$

are specified.

The procedure for investigating equilibria and possible limit cycles is as follows.

Step 1: Solve the equilibrium cubic

Solve

$$\boxed{ax^3 + (d-b)x^2 - (c+I) = 0.} \tag{65}$$

Let the real roots be

$$x_1^*, x_2^*, x_3^*, \tag{204}$$

where only the real roots are retained.

Step 2: Calculate the corresponding y-coordinates

For each real root,

$$\boxed{y_i^* = c - d(x_i^*)^2.} \tag{66}$$

Hence

$$\boxed{E_i = \left(x_i^*, c - d(x_i^*)^2\right).} \tag{67}$$

Step 3: Calculate the trace

For each equilibrium,

$$\boxed{T_i = -3a(x_i^*)^2 + 2bx_i^* - 1.} \tag{68}$$

Step 4: Calculate the determinant

For each equilibrium,

$$\boxed{D_i = x_i^* \left[3ax_i^* + 2(d-b)\right].} \tag{69}$$

Step 5: Classify the equilibrium

If

$$D_i < 0, \tag{205}$$

then E_i is a saddle.

If

$$D_i > 0, \qquad T_i < 0, \tag{206}$$

then E_i is stable.

If

$$D_i > 0, \qquad T_i > 0, \tag{207}$$

then E_i is unstable.

Step 6: Search for Hopf points

Solve

$$\boxed{3ax^2 - 2bx + 1 = 0.} \tag{70}$$

The candidate values are

$$\boxed{x_H^{\pm} = \frac{b \pm \sqrt{b^2 - 3a}}{3a}.} \tag{71}$$

Retain only those candidates for which

$$\boxed{D_H^{\pm} > 0.} \tag{72}$$

Then calculate

$$\boxed{I_H^{\pm} = a(x_H^{\pm})^3 + (d-b)(x_H^{\pm})^2 - c.} \tag{73}$$

Finally, to establish a genuine generic Hopf bifurcation, calculate the first Lyapunov coefficient and verify

$$\boxed{l_1 \neq 0.} \tag{74}$$

30 19. MATLAB Implementation

The following MATLAB code computes the finite equilibria, their Jacobians, eigenvalues, trace, determinant, and basic classification.

```
clear;
clc;

%% Parameters
a = 1;
b = 2;
c = 1;
d = 1;
I = 1;

%% Equilibrium equation
% a*x^3 + (d-b)*x^2 - (c+I) = 0

coef = [a, d-b, 0, -(c+I)];

xroots = roots(coef);

tol = 1e-10;

fprintf('============================================\n');
fprintf('Finite equilibria\n');
fprintf('============================================\n');

count = 0;

for k = 1:length(xroots)

    if abs(imag(xroots(k))) < tol
```

```
count = count + 1;

xstar = real(xroots(k));
ystar = c - d*xstar^2;

%% Jacobian
J = [ -3*a*xstar^2 + 2*b*xstar,  1;
      -2*d*xstar,               -1 ];

%% Eigenvalues
eigJ = eig(J);

%% Trace and determinant
T = trace(J);
D = det(J);

fprintf('\nEquilibrium %d\n', count);
fprintf('x* = %.12f\n', xstar);
fprintf('y* = %.12f\n', ystar);

fprintf('Trace = %.12f\n', T);
fprintf('Determinant = %.12f\n', D);

fprintf('Eigenvalues:\n');
disp(eigJ);

%% Classification
if D < 0

    fprintf('Classification: Saddle\n');

elseif D > 0 && T < 0

    if T^2 - 4*D > 0
        fprintf('Classification: Stable node\n');
    elseif T^2 - 4*D < 0
        fprintf('Classification: Stable focus\n');
    else
        fprintf('Classification: Degenerate stable equilibrium\n');
    end

elseif D > 0 && T > 0

    if T^2 - 4*D > 0
        fprintf('Classification: Unstable node\n');
    elseif T^2 - 4*D < 0
        fprintf('Classification: Unstable focus\n');
    else
        fprintf('Classification: Degenerate unstable equilibrium\n');
    end
```

```
        elseif D > 0 && abs(T) < tol

            fprintf('Hopf candidate: T = 0 and D > 0\n');
            fprintf('Hopf frequency = %.12f\n', sqrt(D));

        else

            fprintf('Classification: Nonhyperbolic / degenerate\n');

        end
    end
end

%% Cubic discriminant
B = d - b;
S = c + I;

Disc = 4*B^3*S - 27*a^2*S^2;

fprintf('\n=============================================\n');
fprintf('Cubic discriminant\n');
fprintf('=============================================\n');

fprintf('Discriminant = %.12f\n', Disc);

if Disc > tol
    fprintf('Three distinct real equilibria.\n');
elseif abs(Disc) <= tol
    fprintf('Multiple equilibrium(s).\n');
else
    fprintf('One real equilibrium.\n');
end

%% Hopf candidates
fprintf('\n=============================================\n');
fprintf('Hopf candidates\n');
fprintf('=============================================\n');

if b^2 - 3*a >= 0

    xH1 = (b + sqrt(b^2 - 3*a))/(3*a);
    xH2 = (b - sqrt(b^2 - 3*a))/(3*a);

    xH = [xH1, xH2];

    for k = 1:2

        xh = xH(k);

        DH = xh*(3*a*xh + 2*(d-b));
```

```
        IH = a*xh^3 + (d-b)*xh^2 - c;

        fprintf('\nHopf candidate %d\n', k);
        fprintf('x_H = %.12f\n', xh);
        fprintf('D_H = %.12f\n', DH);
        fprintf('I_H = %.12f\n', IH);

        if DH > 0
            fprintf('Linear Hopf condition satisfied: D_H > 0.\n');
            fprintf('omega_H = %.12f\n', sqrt(DH));
        else
            fprintf('Not a Hopf point because D_H <= 0.\n');
        end
    end

else

    fprintf('No real Hopf candidates because b^2 - 3*a < 0.\n');

end

%% Divergence test
if a > 0

    maxDiv = (b^2 - 3*a)/(3*a);

    fprintf('\n=============================================\n');
    fprintf('Divergence criterion\n');
    fprintf('=============================================\n');

    fprintf('Maximum divergence = %.12f\n', maxDiv);

    if maxDiv <= tol
        fprintf('Divergence <= 0 for all x.\n');
        fprintf('No planar limit cycle by Bendixson criterion.\n');
    else
        fprintf('Divergence changes sign or can be positive.\n');
        fprintf('Bendixson criterion does not rule out limit cycles.\n');
    end

end
```

Summary of the Main Formulas

For the two-dimensional Hindmarsh–Rose-type model

$$\boxed{\begin{aligned}\dot{x} &= y - ax^3 + bx^2 + I,\\ \dot{y} &= c - dx^2 - y,\end{aligned}} \tag{208}$$

the main formulas for the finite-equilibrium and limit-cycle analysis are summarized below.

Equilibrium equation

$$\boxed{ax^3 + (d-b)x^2 - (c+I) = 0.} \tag{209}$$

y-coordinate

$$\boxed{y^* = c - d(x^*)^2.} \tag{210}$$

Cubic discriminant

$$\boxed{\mathcal{D} = 4(d-b)^3(c+I) - 27a^2(c+I)^2.} \tag{211}$$

Jacobian

$$\boxed{J^* = \begin{pmatrix} -3a(x^*)^2 + 2bx^* & 1 \\ -2dx^* & -1 \end{pmatrix}.} \tag{212}$$

Trace

$$\boxed{T = -3a(x^*)^2 + 2bx^* - 1.} \tag{213}$$

Determinant

$$\boxed{D = x^* \left[3ax^* + 2(d-b)\right].} \tag{214}$$

Eigenvalues

$$\boxed{\lambda_\pm = \frac{1}{2}\left[T \pm \sqrt{T^2 - 4D}\right].} \tag{215}$$

Hopf condition

$$\boxed{T = 0, \qquad D > 0.} \tag{216}$$

Equivalently,

$$\boxed{3ax_H^2 - 2bx_H + 1 = 0.} \tag{217}$$

Hopf equilibrium coordinates

$$\boxed{x_H^\pm = \frac{b \pm \sqrt{b^2 - 3a}}{3a}.} \tag{218}$$

Hopf parameter values

$$\boxed{I_H^\pm = a(x_H^\pm)^3 + (d-b)(x_H^\pm)^2 - c.} \tag{219}$$

Hopf frequency

$$\boxed{\omega_H = \sqrt{D_H}.} \tag{220}$$

No-limit-cycle condition

For

$$a > 0, \qquad b^2 \le 3a, \tag{221}$$

we have

$$\boxed{\nabla \cdot F \le 0 \quad \text{everywhere,}} \tag{222}$$

and therefore, by Bendixson's criterion,

$$\boxed{\text{there is no nonconstant periodic orbit in the plane.}} \tag{223}$$

ooooooooooooooo

31 The two-dimensional Hindmarsh–Rose model and its reduced parameterization

We begin with the two-dimensional Hindmarsh–Rose-type system

$$\boxed{\begin{aligned} \dot{x} &= y - ax^3 + bx^2 + I, \\ \dot{y} &= c - dx^2 - y, \end{aligned}} \tag{1}$$

where x and y denote the fast state variables and a, b, c, d, I are real parameters. In the neuronal interpretation, x represents the membrane potential, while y is a fast recovery (or gating) variable. The parameter I represents the externally applied input current, whereas a, b, c, and d are intrinsic parameters governing the nonlinear activation and recovery mechanisms.

The system in (**??**) contains five parameters. Although retaining all five parameters is useful when describing the general model, a direct analytical treatment of the full five-parameter family is unnecessarily cumbersome for the qualitative analysis pursued in this work, particularly for the determination of equilibria, Hopf bifurcations, periodic orbits, and the global phase portrait. Accordingly, we introduce a mathematically justified reduction of the parameter space while preserving the main biological interpretation of the model.

31.1 Normalization of the cubic coefficient

We first note that, for the case

$$a > 0, \tag{224}$$

the coefficient a can be normalized by an appropriate rescaling of the state variables. Indeed, introduce

$$X = \sqrt{a}\, x, \qquad Y = \sqrt{a}\, (y - c). \tag{2}$$

Equivalently,

$$x = \frac{X}{\sqrt{a}}, \qquad y = c + \frac{Y}{\sqrt{a}}. \tag{225}$$

Since the transformation is a smooth invertible change of coordinates for $a > 0$, it preserves the qualitative phase-space structure up to a smooth coordinate transformation.

Substituting (**??**) into (**??**), we obtain

$$\dot{X} = Y - X^3 + \frac{b}{\sqrt{a}} X^2 + \sqrt{a}(c + I), \tag{3}$$

and

$$\dot{Y} = -\frac{d}{\sqrt{a}} X^2 - Y. \tag{4}$$

Thus, if we introduce the effective parameters

$$\beta = \frac{b}{\sqrt{a}}, \qquad \delta = \frac{d}{\sqrt{a}}, \qquad \mu = \sqrt{a}(c + I), \tag{5}$$

the system takes the normalized form

$$\boxed{\begin{aligned} \dot{X} &= Y - X^3 + \beta X^2 + \mu, \\ \dot{Y} &= -\delta X^2 - Y. \end{aligned}} \tag{6}$$

This transformation shows that, for the qualitative analysis of the two-dimensional autonomous system, the original five parameters can be represented by three effective dimensionless combinations, namely β, δ, and μ. In particular, the normalization of the cubic coefficient is not an additional biological assumption; rather, it corresponds to a choice of scales for the state variables.

For notational simplicity, in what follows we relabel

$$(X, Y, \beta, \delta, \mu) \longmapsto (x, y, b, d, I),$$

so that the normalized system is written as

$$\boxed{\begin{aligned} \dot{x} &= y - x^3 + bx^2 + I, \\ \dot{y} &= -dx^2 - y. \end{aligned}} \tag{7}$$

31.2 Biological interpretation of the normalization

The normalization above should be understood as a nondimensionalization rather than as the biological assertion that the original cubic coefficient is physically equal to one. The coefficient a determines the scale of the cubic nonlinear contribution in the fast voltage equation. By rescaling the state variables, its numerical value can be absorbed into the definition of the dimensionless voltage and recovery variables.

Similarly, the translation

$$Y = \sqrt{a}\,(y - c) \tag{226}$$

centers the recovery variable around the constant offset c. This does not imply that the intrinsic parameter c is biologically absent. Rather, its constant contribution is absorbed into the choice of the origin of the y-coordinate. In the transformed first equation, c and the applied current I occur through the effective combination

$$\mu = \sqrt{a}(c + I). \tag{227}$$

Hence, from the viewpoint of the phase portrait of the reduced two-dimensional autonomous system, c and I do not provide two independent directions in the equilibrium equation.

Nevertheless, we retain the conventional biological interpretation of I as the externally applied current. Therefore, rather than regarding the reduction as an elimination of the biological meaning of c, we retain a biologically meaningful baseline value for c when relating the reduced system to the standard Hindmarsh–Rose parameterization.

31.3 Choice of fixed parameter values

For the subsequent analytical and numerical investigation, we fix

$$\boxed{a = 1, \qquad c = 1,} \tag{8}$$

and retain

$$\boxed{b, \qquad d, \qquad I} \tag{228}$$

as the remaining parameters.

This choice has two complementary justifications. First, the normalization $a = 1$ is mathematically natural because the coefficient of the cubic term can be absorbed by a rescaling of the state variables when $a > 0$, as demonstrated above. Second, the choice $c = 1$ provides a conventional dimensionless baseline for the recovery equation and allows the reduced system to remain close to the standard nondimensional Hindmarsh–Rose formulation.

Consequently, the model considered in the remainder of the analysis is

$$\boxed{\begin{aligned} \dot{x} &= y - x^3 + bx^2 + I, \\ \dot{y} &= 1 - dx^2 - y, \end{aligned}} \tag{9}$$

where

$$\boxed{(b, d, I) \in \mathbb{R}^3} \tag{229}$$

are treated as the effective parameters.

The parameter b controls the quadratic contribution to the fast nonlinear voltage equation and consequently influences the geometry of the cubic nullcline. The parameter d determines the strength of the quadratic contribution in the recovery equation and therefore affects the geometry of the recovery nullcline. Finally, I retains the interpretation of an externally applied input current and is therefore particularly natural as a principal bifurcation parameter.

31.4 Why this parameterization is suitable for the present analysis

The reduction from five parameters to the three-parameter family (**??**) is advantageous for the analysis of limit cycles and global phase portraits. In particular, the equilibrium equation becomes

$$\boxed{x^3 + (d - b)x^2 - (1 + I) = 0,} \tag{10}$$

which is a one-variable cubic equation. Thus the number and location of finite equilibria can be determined explicitly in terms of (b, d, I).

Moreover, the Jacobian matrix at an equilibrium $E^* = (x^*, y^*)$ is

$$J(E^*) = \begin{pmatrix} -3(x^*)^2 + 2bx^* & 1 \\ -2dx^* & -1 \end{pmatrix}. \tag{11}$$

Consequently,

$$\boxed{\operatorname{tr} J(E^*) = -3(x^*)^2 + 2bx^* - 1,} \tag{12}$$

and

$$\boxed{\det J(E^*) = x^* \left[3x^* + 2(d - b)\right].} \tag{13}$$

The Hopf condition is therefore given by

$$3(x_H)^2 - 2bx_H + 1 = 0, \tag{14}$$

and hence

$$\boxed{x_H^{\pm} = \frac{b \pm \sqrt{b^2 - 3}}{3}.} \tag{15}$$

Thus the existence of real Hopf candidates requires

$$\boxed{b^2 \geq 3.} \tag{16}$$

The corresponding values of the input current are obtained from the equilibrium equation:

$$\boxed{I_H^{\pm} = (x_H^{\pm})^3 + (d-b)(x_H^{\pm})^2 - 1.} \tag{17}$$

Therefore, the reduced three-parameter system retains sufficient freedom to investigate equilibrium multiplicity, Hopf bifurcations, periodic orbits, and the associated global phase portraits, while avoiding the substantial algebraic redundancy of the original five-parameter representation.

31.5 Two-parameter global study

For a detailed two-parameter bifurcation and global phase-plane analysis, one may subsequently fix the intrinsic parameter d at the conventional baseline value

$$\boxed{d = 5,} \tag{18}$$

and regard

$$\boxed{(b, I)} \tag{230}$$

as the principal bifurcation parameters. The resulting family is

$$\boxed{\begin{aligned} \dot{x} &= y - x^3 + bx^2 + I, \\ \dot{y} &= 1 - 5x^2 - y. \end{aligned}} \tag{19}$$

The value $d = 5$ is not introduced as a mathematical normalization; rather, it is adopted as a biologically motivated baseline choice consistent with the commonly used dimensionless Hindmarsh–Rose parameterization. In this two-parameter setting, b controls the shape of the fast nonlinear response, while I serves as the externally controllable input current.

Accordingly, the analysis may be organized in two stages. First, the three-parameter family (**??**) is used for the general analytical derivation of equilibria, local stability and bifurcation conditions. Second, the biologically motivated choice $d = 5$ is employed for the detailed exploration of the (b, I) parameter plane, including limit cycles and global phase portraits.

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Simplified 2D Hindmarsh–Rose Model: Finite Singular Points and Bifurcation Analysis

32 The Simplified Model

The general 2D Hindmarsh–Rose-type system is given by:

$$\begin{aligned} \dot{x} &= y - ax^3 + bx^2 + I, \\ \dot{y} &= c - dx^2 - y, \end{aligned} \tag{231}$$

where x represents the membrane potential, y is the recovery variable, and I is the external input current. The parameters a, b, c, d control the intrinsic nonlinearities.

To obtain a mathematically tractable model for global phase-portrait analysis, we fix the parameters as follows:

- $a = 1$
- $c = 1$
- $b = d = k$ (where k is a fixed positive constant, biologically motivated by the standard Hindmarsh–Rose baseline $k = 5$)

The system reduces to:

$$\begin{aligned} \dot{x} &= y - x^3 + kx^2 + I, \\ \dot{y} &= 1 - kx^2 - y. \end{aligned} \tag{S}$$

Here, I is treated as the primary bifurcation parameter.

33 Finite Singular Points

33.1 Solving for the Equilibria

Set $\dot{x} = 0$ and $\dot{y} = 0$. From $\dot{y} = 0$, we obtain:

$$y = 1 - kx^2 \tag{232}$$

Substitute this into $\dot{x} = 0$:

$$(1 - kx^2) - x^3 + kx^2 + I = 0 \tag{233}$$

The kx^2 terms cancel out perfectly, yielding:

$$x^3 = I + 1 \tag{234}$$

For $I > -1$, there exists a **unique real equilibrium**:

$$x^* = \sqrt[3]{I+1}, \qquad y^* = 1 - k(I+1)^{2/3}. \tag{235}$$

This avoids the intractable cubic radicals associated with the general model.

33.2 The Jacobian Matrix

Let $P(x, y) = y - x^3 + kx^2 + I$ and $Q(x, y) = 1 - kx^2 - y$. The Jacobian matrix is:

$$J(x, y) = \begin{pmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{pmatrix} = \begin{pmatrix} -3x^2 + 2kx & 1 \\ -2kx & -1 \end{pmatrix}. \tag{236}$$

Evaluated at the equilibrium (x^*, y^*), and letting $u = x^* = (I + 1)^{1/3}$, we have:

$$J^* = \begin{pmatrix} -3u^2 + 2ku & 1 \\ -2ku & -1 \end{pmatrix}. \tag{237}$$

33.3 Trace and Determinant

The trace T and determinant D of J^* are:

$$T = \mathrm{tr}(J^*) = -3u^2 + 2ku - 1, \tag{238}$$

$$D = \det(J^*) = (-3u^2 + 2ku)(-1) - (1)(-2ku) = 3u^2 - 2ku + 2ku = 3u^2. \tag{239}$$

Notice that $D = 3u^2 > 0$ for all $I > -1$. Therefore, the equilibrium is **never a saddle**. It is either a node or a focus, and its stability is determined entirely by the sign of the trace T.

33.4 Classification of the Singular Point

- If $T^2 - 4D > 0$: Node (stable if $T < 0$, unstable if $T > 0$).
- If $T^2 - 4D < 0$: Focus (stable if $T < 0$, unstable if $T > 0$).
- If $T = 0$ and $D > 0$: Center (linearization) or weak focus.

34 Hopf Bifurcation Analysis

A Hopf bifurcation occurs when $T = 0$ and $D > 0$.

$$T = -3u^2 + 2ku - 1 = 0 \implies 3u^2 - 2ku + 1 = 0. \tag{240}$$

Solving for u:

$$u = \frac{2k \pm \sqrt{4k^2 - 12}}{6} = \frac{k \pm \sqrt{k^2 - 3}}{3}. \tag{241}$$

For real roots, we require $k^2 \geq 3 \implies |k| \geq \sqrt{3}$.

Adopting the biologically standard $k = 5$, we obtain:

$$u = \frac{5 \pm \sqrt{25 - 3}}{3} = \frac{5 \pm \sqrt{22}}{3}. \tag{242}$$

Since $5 > \sqrt{22}$, both roots are positive. The corresponding critical current values are:

$$I_{\mathrm{Hopf}} = u^3 - 1 = \left(\frac{5 \pm \sqrt{22}}{3} \right)^3 - 1. \tag{243}$$

This yields two explicit Hopf bifurcation values for I, allowing for a clean analytical study of the transition from a stable equilibrium to limit cycles (spiking behavior) and back.

Why This Case is Ideal for a New Paper

1. **Tractable Algebra:** The unique equilibrium $x^* = \sqrt[3]{I+1}$ is explicit for $I > -1$, eliminating the need to solve a general cubic.
2. **Clean Trace/Determinant:** The determinant $D = 3u^2$ is always positive, so no saddle points exist in the biological region. Stability is governed entirely by the trace T.
3. **Explicit Hopf Bifurcations:** Two explicit Hopf bifurcation points in terms of I provide a clear route to studying limit cycles.
4. **Poincaré Compactification:** The highest-degree terms ($-x^3$ and $-kx^2$) preserve the essential nonlinear structure, making the behavior at infinity tractable while retaining exact control over the finite singular point.
5. **Biological Meaning:** Setting $b = d$ is a mathematical normalization that preserves the fast-slow structure and nonlinear spiking behavior of the HR neuron.

35 The Simplified Model

We study the system

$$\begin{aligned} \dot{x} &= y - x^3 + kx^2 + I, \\ \dot{y} &= 1 - kx^2 - y, \end{aligned} \tag{S}$$

where k is a fixed positive constant (e.g., $k = 5$) and I is the primary bifurcation parameter. The polynomial degree is $n = 3$.

36 Poincaré Compactification

To study the behaviour at infinity, we use the Poincaré compactification. We consider the local charts U_1 and U_2.

36.1 Local Chart U_1

In the local chart U_1, we set

$$x = \frac{1}{u}, \qquad y = \frac{v}{u}.$$

After the change of variables and a time rescaling $d\tau = dt/u^{n-1} = dt/u^2$, system (S) becomes

$$\begin{aligned} \dot{u} &= -u^3 v + u - ku^2 - Iu^4, \\ \dot{v} &= u^3 - ku - u^2 v - u^2 v^2 + v - kuv - Iu^3 v. \end{aligned} \tag{U1}$$

The line at infinity in this chart is $u = 0$. Setting $u = 0$ in (U1) yields

$$\dot{u} = 0, \qquad \dot{v} = v.$$

Thus the only infinite singular point in U_1 is

$$(u, v) = (0, 0).$$

The Jacobian matrix of (U1) at $(0, 0)$ is

$$J_{U_1}(0,0) = \begin{pmatrix} 1 & 0 \\ -k & 1 \end{pmatrix}.$$

Its eigenvalues are both 1. Therefore, the infinite singular point $(0, 0)$ in U_1 is a **hyperbolic unstable node** (a source) on the equator. This point corresponds to the direction of the x-axis at infinity.

36.2 Local Chart U_2

In the local chart U_2, we set

$$x = \frac{u}{v}, \qquad y = \frac{1}{v}.$$

After the change of variables and a time rescaling $d\tau = dt/v^{n-1} = dt/v^2$, system (S) becomes

$$\begin{aligned} \dot{u} &= -u^3 + ku^3v + ku^2v + uv^2 - uv^3 + v^2 + Iv^3, \\ \dot{v} &= v^3(1 - v + ku^2). \end{aligned} \tag{U2}$$

The line at infinity in this chart is $v = 0$. Setting $v = 0$ in (U2) yields

$$\dot{u} = -u^3, \qquad \dot{v} = 0.$$

Thus the only infinite singular point in U_2 is

$$(u, v) = (0, 0).$$

The Jacobian matrix of (U2) at $(0, 0)$ is identically zero:

$$J_{U_2}(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence $(0, 0)$ is a **linearly zero singular point**. Its local phase portrait must be determined by the blow-up technique. The leading terms near the origin are

$$\dot{u} \approx -u^3 + v^2, \qquad \dot{v} \approx v^3.$$

This is a quasi-homogeneous singular point of type $(3, 2)$. A standard blow-up analysis (e.g., using weights $u = r^2X$, $v = r^3Y$) shows that the origin is formed by a combination of hyperbolic and parabolic sectors. In particular, the flow on the equator $v = 0$ satisfies $\dot{u} = -u^3$, so $u = 0$ is a stable equilibrium along the equator. Off the equator, $\dot{v} = v^3$, so the flow moves away from the equator. The detailed sector decomposition yields **two hyperbolic sectors and two parabolic sectors**, similar to the even-γ case in the Higgins–Selkov system. This point corresponds to the direction of the y-axis at infinity.

37 Summary of Infinite Singular Points

- In chart U_1: $(0, 0)$ is a hyperbolic unstable node (source) on the equator.
- In chart U_2: $(0, 0)$ is a linearly zero singular point, which after blow-up consists of two hyperbolic and two parabolic sectors.

These two points are the only infinite singular points on the Poincaré sphere. The qualitative behaviour at infinity is independent of the parameter I for $I > -1$, since the leading terms are governed by the fixed cubic and quadratic nonlinearities.